

Lecture 1

Introduction to Econometrics

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Recommended textbooks

Main:

- Gujarati, D.N., and Porter, D.C., (2009) Basic Econometrics, Fifth Edition, McGraw-Hill

Supplementary:

- Wooldridge, G., (2013) Introductory Econometrics, 5th ed
- and others available in the Library

Computer software

Main:

- Stata

Supplementary:

- Eviews
- Rstudio
- Excel

"... the application of mathematical statistics to economic data to lend empirical support to models constructed by mathematical economics and to obtain numerical estimates ..."
(Samuelson et al., Econometrica, 1954)

The Nature of Econometrics

- What is Econometrics?
 - Econometrics - use of statistical methods to analyze economic data
 - Econometricians typically analyze non-experimental data
- Typical goals of econometric analysis
 - Estimating relationships between economic variables
 - Testing economic theories and hypothesis
 - Forecasting economic variables
 - Evaluating and implementing government and business policy

What is econometrics?

Econometrics, the result of a certain outlook on the role of economics, consists of the application of mathematical statistics to economic data to lend empirical support to the models constructed by mathematical economics and to obtain numerical results.
(Gerhard Tintner 1968)

Finding the set of assumptions that are both sufficiently specific and sufficiently realistic to allow him to take the best possible advantage of the data available to him (E. Malinvaud 1966)

Conjunction of economic theory and actual measurements, using the theory and technique of statistical inference as a bridge pier.

What is econometrics?

Mathematical economics is to express economic theory in mathematical form (equations) without regard to measurability.

Economic statistics - collecting, processing, and presenting economic data in the form of charts and tables. – but not testing economic theories.

Econometrics - empirical verification of economic theories

Econometric methodology

- 1 Statement of **theory or hypothesis**
- 2 Specification of the **mathematical model** of the theory
- 3 Specification of the **statistical, or econometric, model**
- 4 Obtaining the **data**
- 5 Estimation of the **parameters of the econometric model**
- 6 **Hypothesis testing**
- 7 **Forecasting or prediction**
- 8 Using the model for control or **policy purposes**.

Statement of theory or hypothesis

What happens if government devaluates the currency exchange rate on exports?

Theory says:

$$Exports = f(Exchange\ rate, Foreign\ income, Domestic\ price, Foreign\ price)$$

$$Exports = f(Exchange\ rate)$$

Mathematical model of the theory

$$U = f(C^d, C^f)$$
$$I = P^d C^d + P^f C^f Ex$$

$$C^d = f(I, P^d, P^f, Ex)$$

$$C^f = f(I, P^f, P^d, Ex)$$

Econometric model

Linear or nonlinear models:

$$C_t^f = \alpha + \beta_1 I_t + \beta_2 P_t^f + \beta_3 P_t^d + \beta_4 Ex_t + u_t$$

What if they are nonlinear?

The Nature of Economic Data

Econometric analysis requires data.

Economic data sets:

- Cross-sectional
- Time series
- Pooled
- Panel\Longitudinal

The Nature of Economic Data

Cross-sectional data sets:

- Sample of individuals, households, firms, cities, states, countries, or other units of interest at a given point of time/in a given period
- Cross-sectional observations are more or less independent
- For example, pure random sampling from a population
- Sometimes pure random sampling is violated, e.g. units refuse to respond in surveys, or if sampling is characterized by clustering
- Cross-sectional data typically encountered in applied microeconomics

The Nature of Economic Data

Cross-sectional data set on wages and other characteristics

obsno	wage	educ	exper	female	married
1	3.10	11	2	1	0
2	3.24	12	22	1	1
3	3.00	11	2	0	0
.	.	.	.	.	.
.	.	.	.	.	.

obsno - observation number

wage - hourly wage

educ - years of education

exper - years of experience

female, married - indicator variables

The Nature of Economic Data

Time series data:

- Observations of a variable or several variables over time
- For example, stock prices, money supply, consumer price index, gross domestic product,
- annual homicide rates, automobile sales, ...
- Time series observations are typically serially correlated
- Ordering of observations conveys important information
- Data frequency: daily, weekly, monthly, quarterly, annually, ...
- Typical features of time series: trends and seasonality
- Typical applications: applied macroeconomics and finance

The Nature of Economic Data

Time series data on minimum wages and related variables

obsno	year	avgmin	avgcov	prunemp	prgnp
1	1950	0.20	20.1	15.4	878.7
2	1951	0.21	20.7	16.0	925.0
3	1952	0.23	22.6	14.8	1015.9
.	.	.	.	.	.
.	.	.	.	.	.

avgmin - average minimum wage for the given year

avgcov - average coverage rate

prunemp - unemployment rate

prgnp - gross national product

The Nature of Economic Data

Panel or longitudinal data:

- The same cross-sectional units are followed over time
- Panel data have a cross-sectional and a time series dimension
- Panel data can be used to account for time-invariant unobservables
- Panel data can be used to model lagged responses

Example:

- City crime statistics; each city is observed in two years
- Time-invariant unobserved city characteristics may be modeled
- Effect of police on crime rates may exhibit time lag

The Nature of Economic Data

Two-year panel data on city crime

obsno	city	year	murders	population	unem	police
1	1	1986	5	350,000	8.7	440
2	1	1990	8	359,200	7.2	471
3	2	1986	2	64,300	5.4	75
4	2	1990	1	65,100	5.5	75
.	.	.	.	.	.	.
.	.	.	.	.	.	.

Each city has two time series observations.

Model specification

- the equation(s) - variables and functional form
- a priori restrictions on parameters
- stochastic assumptions about disturbance term

Estimation of the parameters

$$Export_t^f = 0.02 + 0.2I_t - 0.12P_t^f + 0.23P_t^d + 0.05Ex + u_t$$

All variables in growth form (%)

Types of model

- simple bivariate linear
- non-linear bivariate
- multiple regression
- dynamic
- simultaneous equation
- panel
- other (e.g. logit and probit)

Assumptions about u

- mean zero
- constant variance
- independent between observations
- independent of the X variable

The role of the disturbance term

Reasons for the disturbance:

- random nature of human behaviour
- omitted influences on Y
- errors of measurement
- non-linearity
- others

Reference

Gujarati, D.N., and Porter, D.C., (2009), Basic Econometrics, 5th edition. Chapter 1 and 2